

TECHHUB ELECTRONICS

DIGITAL TRANSFORMATION ROADMAP

A 36-month plan for a modern website, connected online/store services and better warehouse visibility

BCS CAPSTONE PROJECT

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Executive summary

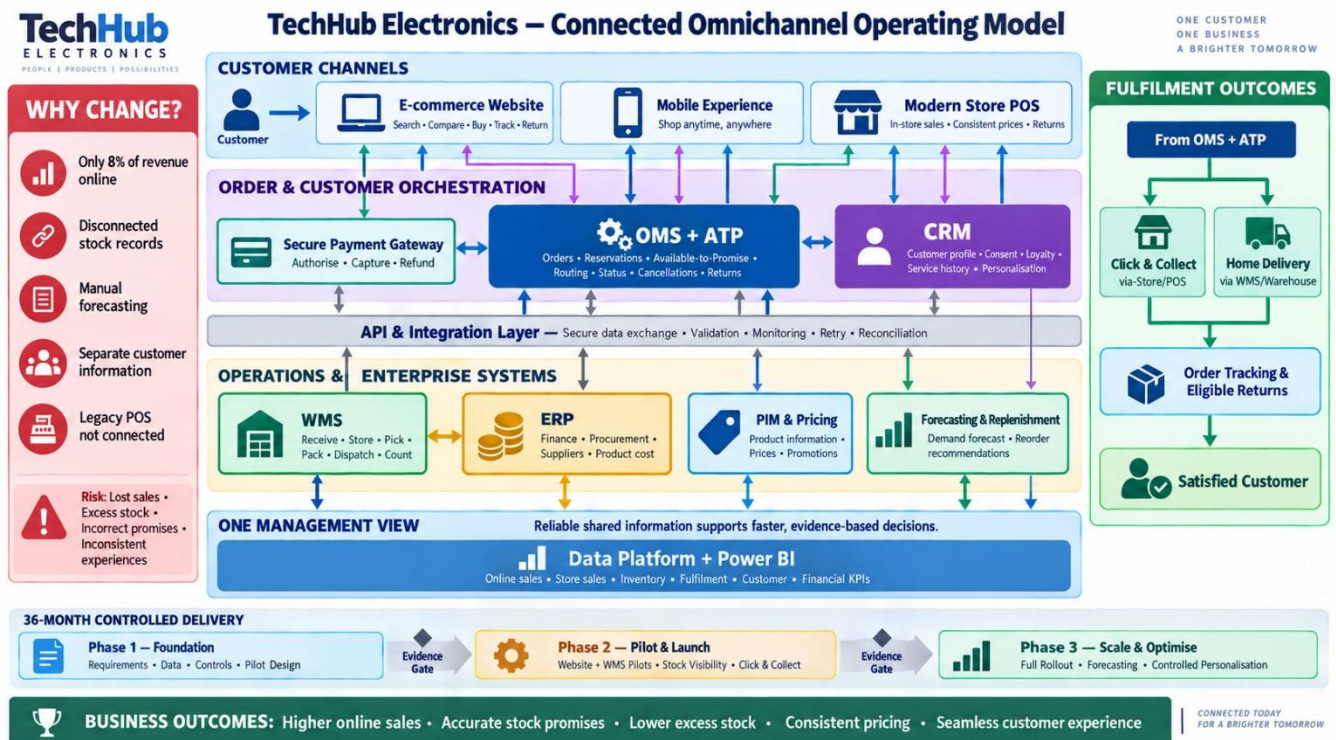
TechHub Electronics has a strong brand and a national store network, but its digital services and operations have not kept pace with customers. Only 8% of revenue comes from online sales. The website has limited shopping features, store and warehouse stock records do not always agree, demand forecasting is manual, customer information is stored separately and the older point-of-sale (POS) systems are not connected to the website.

These problems can cause lost sales, excess stock, incorrect delivery promises, inconsistent prices and different experiences online and in stores. The need for change is clear. Australia Post reported that 82% of Australian households shopped online in 2025.

The recommended strategy starts with a modern, mobile-friendly website connected to stores and the warehouse. Customers will be able to search and compare products, see reliable stock, pay securely, choose delivery or click-and-collect, track orders and manage eligible returns. This connected experience is called omnichannel: customers can move between online and store services without starting the process again.

A warehouse management system (WMS) will control receiving, storage, picking, packing, dispatch and stock counts using scanners. An order management system (OMS) will manage each order from purchase to delivery or collection. Its available-to-promise (ATP) calculation will combine warehouse and store stock, subtract reservations and safety stock, and show what TechHub can safely promise to customers.

The work will be delivered over 36 months. Phase 1 confirms requirements, data, controls and pilot designs. Phase 2 launches the website and WMS pilots, introduces reliable stock information and tests click-and-collect. Phase 3 completes the rollout, improves forecasting and introduces carefully controlled personalisation. Pilots and evidence-based decision points will reduce disruption to stores and warehouse operations.



Executive recommendations

Priority	Recommendation	Business outcome
1	Rebuild the website so it is fast, accessible, mobile-friendly and easy to use.	Increase online sales, help customers find products and make checkout easier.
2	Implement an OMS with ATP stock checks, reservations and one order status.	Provide reliable stock promises, prevent overselling and support delivery and click-and-collect.
3	Implement a WMS with handheld scanning and serial-number tracking.	Improve warehouse stock accuracy, speed, visibility and order reliability.
4	Connect systems through secure APIs and assign owners for product, price, customer and stock data.	Create one trusted source of information and reduce manual checking.
5	Connect store POS, customer service, reporting and security, supported by staff training.	Make the change practical, secure, measurable and sustainable.

Expected outcomes and investment

- Increase digital revenue contribution from 8% to a planning target of 20% by Month 36, subject to baseline validation and commercial testing.
- Show customers and staff reliable stock availability across warehouse and store locations.
- Raise warehouse inventory accuracy to at least 98% and order accuracy to at least 99%, with final targets confirmed by pilot measurement.
- Reduce stockouts by 40%, improve inventory turnover by 15% and reduce warehouse fulfilment cycle time by 30% relative to validated baselines.
- Invest \$18.6 million during the three-year transformation, within the supplied \$15–20 million envelope.
- Target a three-year return on investment (ROI) of 14.5%, payback around Month 33 and a five-year ROI of approximately 145%. These are estimates.

1 Current state assessment

TechHub is a \$280 million electronics retailer with 65 metropolitan stores and 850 employees. It sells consumer electronics, gaming products, smart-home devices and mobile accessories. Leadership supports the transformation and has provided a significant budget. However, the website, stores and warehouse still operate as separate channels. Information that is not provided in the case brief is marked for validation rather than presented as fact.

1.1 Business process analysis

Process Area	Current State	Key Issues	Impact Level
Customer Management	Customer information is limited and separated across the website, stores and customer service.	No complete customer record, marketing-consent record or shared service history.	High
Inventory Management	Store and warehouse systems report discrepancies; forecasting and comparisons are manual.	No trusted inventory position or consistent reservations across channels.	Very high
Sales Process	Physical stores dominate; online sales are only 8%; ageing POS and website are disconnected.	Prices, promotions, carts and order history are not consistently shared.	High
Warehouse Fulfilment	The case brief does not show that warehouse work is directed by a system or controlled through scanning.	Limited visibility, task control, problem handling and stock-count accuracy.	Very high
Supply Chain	Demand forecasting is manual and the business responds slowly to changes.	Forecasts are slow to update and stock is not allocated using clear rules.	Very high
Data Management	Systems are disconnected and reporting is limited.	Product, location and customer data does not have one agreed definition or owner.	High

1.2 Technology infrastructure

System/Technology	Description	Limitations	Performance Issues
POS systems	Ageing point-of-sale systems support transactions across the 65-store network.	Disconnected from the website, customer data and cross-channel order processes.	Pricing and product-availability complaints, separate order history and additional manual checking.
Inventory / warehouse systems	Store and warehouse systems hold separate stock records. The case brief does not confirm an existing WMS.	No single view of physical stock, reserved stock or available-to-promise stock. Scanner, bin, serial-number and stock-count controls are not confirmed.	Stock differences, stockouts, excess stock, incorrect customer promises and manual checking.
E-commerce platform	Basic website generating approximately 8% of total revenue.	Limited commerce functionality, fulfilment choice, order tracking, analytics and integration.	Low online contribution and an inconsistent customer experience across website and stores.
Customer data systems	Customer-data collection and reporting are limited.	Customer identity, preferences, consent, purchases and service history are stored separately.	Generic marketing, limited loyalty information and customers repeating information to service staff.
ERP / finance / procurement	The underlying ERP, finance and procurement applications are not identified in the case brief.	Integration capability and master-data ownership cannot yet be confirmed.	Manual forecasting and limited analytics reduce responsiveness and decision speed.
Integration / data platform	The technology environment is disconnected and combined reporting is limited.	No confirmed standard APIs, complete monitoring of system connections.	Different information between systems, delayed reports, manual fixes and extra checking.

1.3 Organisational readiness

Dimension	Working rating	Assessment	Implication
Leadership commitment	4 / 5	Leadership is committed but concerned about disruption.	Use formal decision gates, operational pilots and visible benefit reporting.
Technology skills	2 / 5	Digital skills vary across the 850 employees.	Provide training by job role, on-site support and simple mobile work steps.
Change management capacity	2.5 / 5	The company is changing its management approach, but its ability to deliver a large transformation is not yet proven.	Create a dedicated programme team and local change representatives in stores and the warehouse.
Financial resources	4 / 5	\$15–20m is available over three years.	The programme is affordable if work is delivered in stages and emergency funds are protected.
Cultural factors	3 / 5	The family-owned history may support loyalty, but teams may continue local workarounds.	Design processes with employees and standardise them only after testing.

2 Future state vision

2.1 Business vision statement

Vision: By Month 36, TechHub will operate as one connected electronics retailer. Customers can search, buy, collect, receive, service and return products through supported channels. Employees will use reliable stock, customer and order information.

The website will become TechHub's main digital store, but it will not operate separately. It will connect with physical stores, customer service, the warehouse, delivery providers and business systems. Customers will receive the same product information, prices, stock availability and order status in every supported channel. Employees will use information and tasks designed for their roles instead of checking several systems manually.

2.2 Target operating model

After the transformation, the website, stores, warehouse and support teams will work through one customer-centred model. The following sections cover the five areas required by the template: customer experience, employee experience, operational efficiency, data-driven decision-making and technology integration.

2.2.1 Customer experience

Customers will experience TechHub as one retailer rather than separate website and store businesses. Product information, prices, availability, fulfilment choices, order status and returns policies will remain consistent across supported channels.

Journey stage	Future customer experience	Enabling capability	Operational control
Discover & compare	Fast mobile pages, easy search and filtering, clear specifications, compatibility details and product comparison.	E-commerce website, content and product information system, search tools and analytics.	A named content owner, product-data checks and website-speed monitoring.
Check availability	Show store and delivery availability before checkout, with realistic dates.	OMS available-to-promise (ATP) calculation using WMS and store/POS stock.	Safety stock, reservations, update-time monitoring and alerts when data is unreliable.
Buy & receive	Guest or customer-account checkout; delivery or click-and-collect; consistent promotions and secure payment.	Website, OMS, payment gateway, store tasks and WMS.	Payment Card Industry (PCI DSS) security, order-routing rules, store capacity and service monitoring.
Track & support	One order status, automatic notifications and customer-service access to order details.	OMS order status, customer record and notification service.	Complete status updates, controlled access and a named owner for each customer case.
Return & retain	Return eligible products through supported channels with clear Australian Consumer Law information.	Return process, order history, WMS/store hold area and refund connection.	Serial-number checks, return decisions, refund controls and return-reason reporting.

2.2.2 Employee experience by department

Employee experience will be designed for each department. Every team will receive the information, work steps and decision authority needed for its role. Training will cover both the new systems and the new standard operating procedures.

Department	Future way of working	Primary capabilities	Employee outcome
Retail / store operations	Use one view of products, prices, stock and customer orders. Receive clear tasks for collection, returns and stock changes.	Connected POS and OMS, handheld scanners and customer/order search.	Fewer calls and manual checks, clearer priorities and faster service.
Warehouse operations	Use scanning for receiving, putaway, replenishment, picking, packing, dispatch, stock counts and returns.	WMS, handheld devices, bin locations, serial-number tracking and dashboards.	More accurate stock, less paperwork and earlier warning of problems.
Supply chain planning & procurement	Plan purchases and replenishment using reliable demand, supplier and stock information.	Forecasting reports, purchasing system, supplier delivery data and stock visibility.	Less manual comparison, faster purchasing and better stock balance.
Customer service	See the customer's order, delivery, collection and return history in one place.	Customer relationship management (CRM), OMS, notifications and return process.	Faster answers, fewer hand-offs and clear ownership of each case.
Marketing & e-commerce	Manage product content, campaigns, search and promotions using measured results.	E-commerce platform, product information, customer data and website analytics.	Faster content updates, consistent offers and decisions based on evidence.
Finance	Match sales, payments, refunds, stock values and benefits using reliable transaction and KPI data.	Finance-system connection, payment data, business intelligence reports and benefits dashboard.	Less manual reconciliation and clearer reporting of margin, investment and benefits.
Information technology	Monitor systems and connections with clear ownership, security and recovery procedures.	Secure APIs, system monitoring, access control, support tools and security tools.	Earlier fault detection, fewer manual fixes and clearer support responsibility.
Human resources & change	Manage role changes, training, skills, adoption and staffing needs for each rollout stage.	Learning system, change representatives, adoption reports and workforce planning.	Clear roles, targeted training and evidence that employees are ready.
Executive management	Review customer, sales, stock, delivery, risk and benefit measures in one agreed report.	Combined business-intelligence dashboard, governance meetings and Finance-approved benefits.	Faster decisions, clear accountability and controlled investment.

2.2.3 Operational efficiency

- The WMS will direct receiving, storage, replenishment, picking, packing, dispatch and stock counting. This will reduce paper-based warehouse work.
- The OMS will use ATP information to reserve stock and select the best store or warehouse for each order.
- Stores will use standard tasks for collecting products, preparing orders, confirming collection and processing returns.
- Secure system connections and automatic checks will reduce duplicate data entry, manual comparison and unnoticed errors.
- Process owners will monitor speed, accuracy, productivity, cost and unresolved problems across the complete order process.

2.2.4 Data-driven decision-making

- Finance and business owners will agree each KPI, its starting value, target, data source and benefit-calculation method.
- Managers will use combined dashboards for website sales, stock, orders, warehouse work, store fulfilment, customers and financial results.
- Operational teams will receive alerts about problems instead of waiting for manual reports.
- Demand forecasting will be introduced only after the historical data and operating processes are reliable. Planners will approve important stock decisions.
- Website personalisation will use customer-consent controls and measured business results, not website traffic alone.

2.2.5 Technology integration

The website, OMS, WMS, store POS and business systems will share information through secure connections called APIs. Clear data rules, monitoring and named system owners will keep these connections reliable.

Digital commerce and omnichannel capabilities

- A fast, mobile-friendly and accessible website with search, filtering, product comparison, reviews and checkout.
- One central source for product descriptions, specifications, compatibility, warranty, safety and delivery information.
- Reliable store and warehouse availability, stock reservations and realistic delivery or collection dates.
- Click-and-collect, home delivery, order tracking, digital receipts and returns across supported channels.
- Secure debit and credit card payments using a certified payment gateway and Payment Card Industry (PCI DSS) controls.
- Consistent prices and promotions, customer accounts, loyalty information and marketing-consent preferences.

Target connected retail system design

WMS controls warehouse execution; the inventory/OMS layer coordinates availability and promises across all channels.

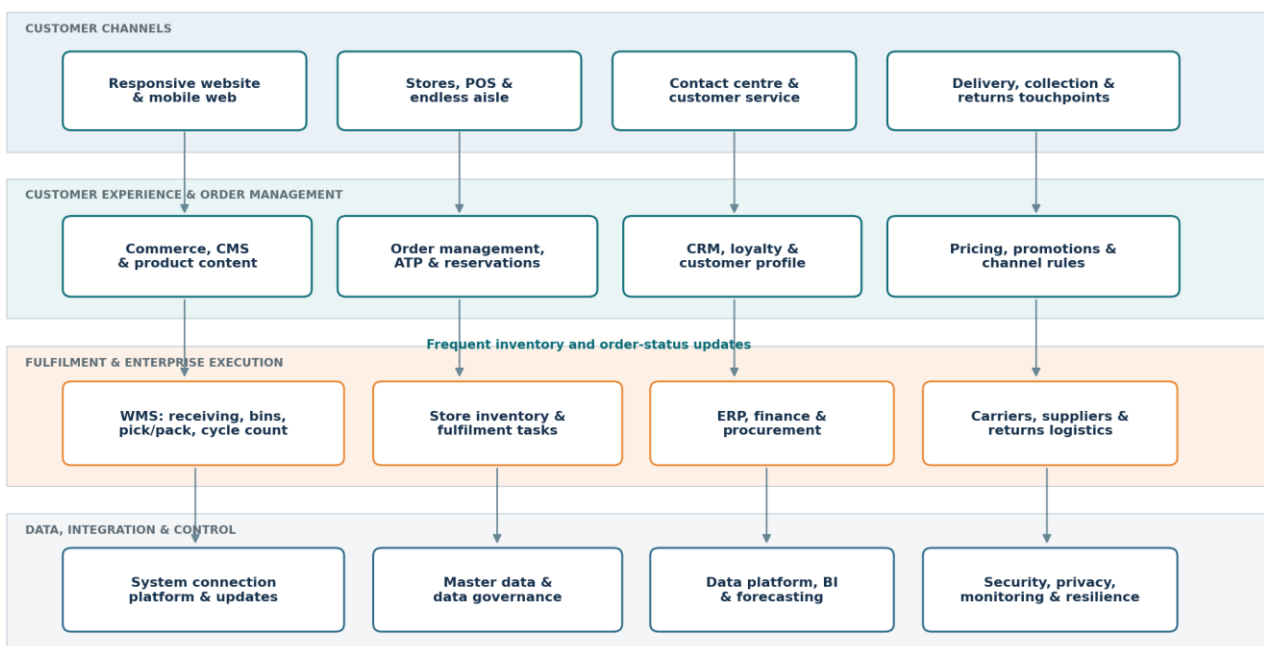


Figure 1. Target connected retail system design (author-developed).

2.3 Success metrics

The table follows the project template. It shows the starting position, three-year target and measurement method for each area. Only the 8% online-revenue figure comes directly from the case brief. All other figures are planning assumptions that must be checked during Months 1–3.

Metric category	Current baseline	3-year target	Measurement method
Customer satisfaction	Validate in Phase 1	Increase by 12 points	Post-purchase, delivery, collection and return journey surveys.
Revenue growth	\$280m annual revenue	At least 8% growth above the validated baseline	Finance revenue and incremental gross-margin reporting.
Operational efficiency	Cycle-time and productivity audit required	Warehouse fulfilment cycle time -30%; pick productivity +20%	WMS timestamps, labour hours and cost-to-serve dashboard.
Digital revenue %	8%	20%	Finance channel revenue reconciled to commerce and order data.
Inventory turnover	Validate in Phase 1	Increase by 15% without reducing service level	Finance cost of goods sold and average inventory.

Metric category	Current baseline	3-year target	Measurement method
Website conversion	Audit required	Increase by 60% relative to baseline	Commerce analytics with bot/internal traffic excluded.
Inventory accuracy	Store/warehouse discrepancies reported	Warehouse ≥98.5%; store ≥97.5%	Cycle-count variance and inventory reconciliation.
Omnichannel service	No established capability	At least 95% of collection orders ready on time	OMS order-status timestamps and exception codes.
Employee adoption	Not currently measured	At least 90% active use and role proficiency	Training completion, skills checks and system-usage reports.

Stakeholder value propositions

Stakeholder	Value proposition	Evidence of value
Customers	Confidence that products, prices and delivery promises are accurate; freedom to move between online and store channels.	Conversion, promise accuracy, collection ready-time, complaints, repeat purchase and satisfaction.
Store employees	One view of stock, customer order and fulfilment tasks with fewer calls and manual checks.	Task completion, search time, rework and employee adoption.
Warehouse employees	Directed scan-based work, bin-level visibility and earlier exception alerts.	Pick rate, order accuracy, cycle-count variance and dock-to-stock time.
Management	Integrated commercial and operational information with traceable benefits.	Margin, inventory turns, service level, digital mix and realised savings.
Suppliers / carriers	Clearer transactions, status updates and connection standards.	Supplier delivery-data quality, delivery problems, lead time and claims.

3 Gap analysis

3.1 Capability gaps

The table compares TechHub's current capability with the level needed after transformation. The scale is 1 = very limited, 2 = partly repeatable, 3 = defined, 4 = measured and connected, and 5 = fully optimised. These scores are estimates and must be checked during Phase 1.

Capability Area	Current Level	Required Level	Gap Description	Priority
Customer analytics	2	4	Customer information is stored separately; there is no complete profile, consent record or cross-channel view.	High
Omnichannel experience	1	4	Customers cannot move smoothly between the website, stores, collection, delivery, support and returns.	Critical
Demand forecasting	1.5	4	Forecasting is manual and lacks reliable demand, inventory and supplier signals.	High
Real-time inventory	1.5	4	Store and warehouse stock figures can differ; there is no reliable available-to-promise calculation.	Critical
Personalisation	1	4	There is no reliable customer profile, consent control or measured recommendation service.	High
Digital commerce & user experience	2	4	The website lacks strong mobile shopping, product search, checkout, secure payment and performance monitoring.	Critical
Omnichannel order management	1	4	There is no single process for order capture, reservation, routing, status, cancellation or return.	Critical
Warehouse execution	2	4	Current WMS capability is unknown. Scanning, bin visibility, directed tasks, stock counts and problem handling must be established.	Critical
Product / price master data	2	4	No confirmed owner or automatic update process across the website, POS and business systems.	High
Integration & monitoring	1.5	4	System connections are unreliable and lack complete error monitoring and recovery.	Critical
Cybersecurity, privacy & resilience	1.5	4	Security and privacy controls must be improved for online shopping, payments, connected systems and customer data.	Critical
Change & support model	2	4	Employee skills vary and TechHub has not yet proven its ability to deliver a transformation of this size.	High

3.2 Technology gaps

No reliable inventory availability: The website, stores and warehouse do not share accurate stock information. Connect the POS, WMS, website and OMS so stock is updated in near real time. Available stock must consider existing orders, reservations and safety stock.

No unified order management: Orders cannot be managed consistently across the website, stores and warehouse. Implement an Order Management System (OMS) to route, collect, cancel, return and track orders through one standard process.

Warehouse activities are not connected to customer orders: Warehouse picking, packing, dispatch and delays are not clearly connected to the delivery information shown to customers. Implement a WMS and share order progress with the OMS and website.

No online card-payment facility: Customers cannot securely pay by debit or credit card through the website. Introduce a secure payment gateway with PCI DSS controls, fraud monitoring, refunds and payment-status updates.

Inconsistent product and pricing information: Product descriptions, specifications, compatibility information, prices and promotions may differ between the website and stores. Introduce a central product information system and assign clear responsibility for product and pricing data.

Systems are poorly connected: Separate systems and manual data transfers can cause delays, duplicate information and unnoticed errors. Use secure APIs to connect systems and introduce automatic retries, data checks and error monitoring.

Limited customer-data and privacy controls: TechHub does not have a reliable process for customer profiles, preferences and marketing consent. Introduce suitable privacy, consent, access and data-retention controls.

No clear digital-service standards: Targets for website speed, availability, accessibility, security and recovery are not established. Define service targets and use automated testing and monitoring.

4 Strategic initiatives

Each initiative was rated by business value, delivery effort, dependencies and risk. The website, OMS/inventory visibility and WMS have the highest priority. However, they still depend on reliable data, system connections, store processes, security and employee training.

4.1 Initiative prioritisation matrix

Initiative	Business Impact	Implementation Effort	Dependencies	Priority Score
1. Digital commerce and website rebuild	Very high	High	Product data, pricing, payments, analytics and accessibility	9.2
2. Omnichannel OMS, ATP and inventory visibility	Very high	Very high	WMS/store feeds, master data, integration and order rules	9.1
3. WMS-enabled warehouse transformation	Very high	Very high	Warehouse redesign, devices, item/location data and interfaces	8.9
4. API integration, master data and monitoring	Very high	High	System design, data ownership, security and source-system access	8.7
5. POS and store-fulfilment enablement	High	High	OMS, store accuracy, network and labour model	8.0
6. Security, privacy, resilience and change programme	High	Medium	Executive sponsorship and cross-stream standards	7.9
7. CRM, loyalty and service integration	High	Medium	Consent, customer identity and order history	7.4
8. BI, demand forecasting and personalisation	High	High	Clean historical data and stable operational processes	7.2

Strategic initiative prioritisation

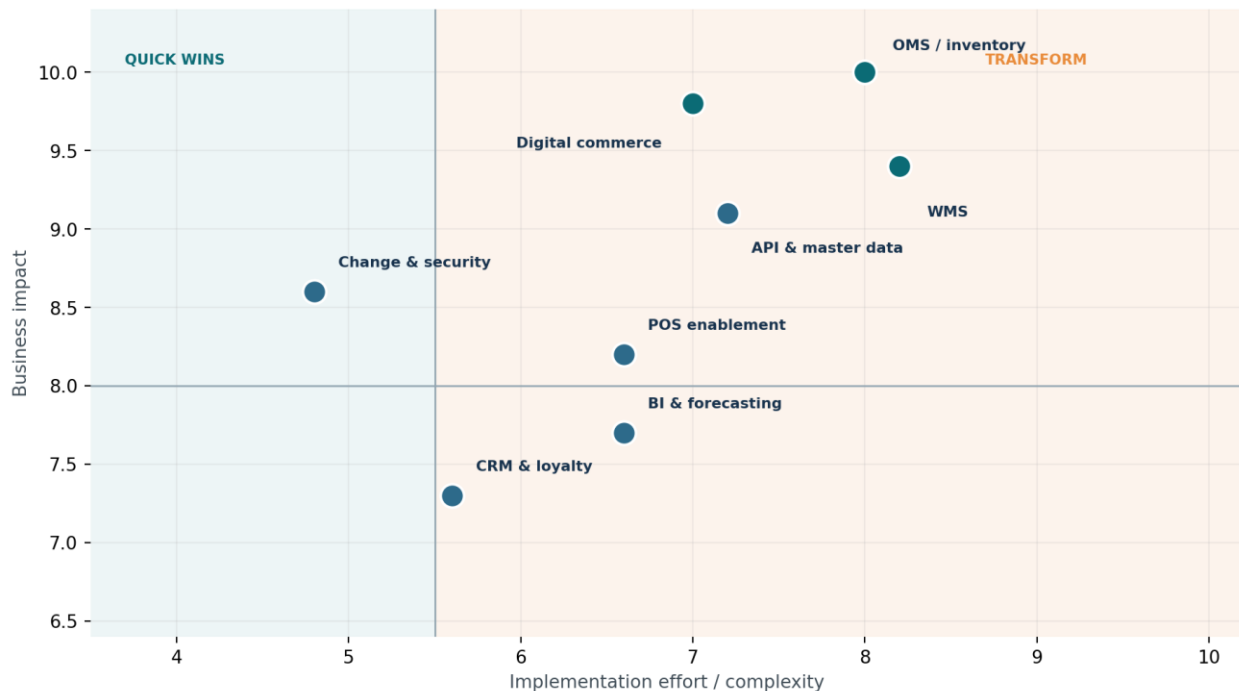


Figure 2. Impact-versus-effort view of the strategic initiative portfolio (author-developed).

4.2 Detailed initiative descriptions

4.2.1 Initiative 1 — Digital commerce and website rebuild

Objective: create a fast, accessible and easy-to-use website with reliable product, price, stock and delivery information.

- Included: mobile-friendly design, product search and comparison, product information, checkout, secure card payment, customer accounts, order tracking, delivery or collection choices, returns information and website reporting.
- Not included in the first release: a separate mobile app, marketplace sales or advanced AI shopping tools. These can be considered after the website is stable.
- Timeline: discovery and design M1–4; build/integration M4–10; controlled release M11–14; optimisation throughout M15–36.
- Success: online revenue reaches 20%, website conversion improves by 60%, key pages meet agreed speed and accessibility standards, and customer satisfaction increases by 12 points by Month 36.
- Indicative three-year allocation: \$3.0m, including experience design, platform implementation, content migration, testing and initial optimisation.

4.2.2 Initiative 2 — Omnichannel OMS, ATP and inventory visibility

Objective: manage every order through one process and provide reliable stock promises by combining store and warehouse availability.

- Capabilities: physical stock, available stock, ATP dates, reservations, safety stock, split orders, click-and-collect, home delivery, cancellations, status updates and returns.
- Business rules: select the best store or warehouse by stock, distance, capacity, delivery time, product restrictions and cost.
- Timeline: design M3–6; build M6–13; pilot M11–18; scale M18–29; optimise M30–36.
- Success: all active locations are included in ATP, at least 99.5% of stock updates are processed, normal updates arrive within five minutes, overselling stays below the agreed limit and at least 95% of collection orders are ready on time.
- Indicative allocation: \$2.5m for the OMS, stock-availability service, order rules and system connections.

4.2.3 Initiative 3 — WMS-enabled warehouse transformation

Objective: use scanning and system-directed tasks to create accurate warehouse operations that support online and store orders.

- Capabilities: supplier delivery notices, receiving, serial-number scanning, directed putaway, bin control, replenishment, group or order picking, packing, dispatch, stock counting, holds, returns and performance dashboards.
- Required support: improved warehouse layout and processes, clean item and location data, reliable Wi-Fi, handheld scanners, printers, labels, system connections, training and on-site support.
- Timeline: observe current work and design the solution in Months 1–5; configure, clean data and connect systems in Months 4–11; run a controlled pilot in Months 11–15; stabilise and roll out in Months 15–24.
- Success: $\geq 98.5\%$ inventory accuracy, $\geq 99.2\%$ order accuracy, fulfilment cycle time -30% , pick productivity $+20\%$ and important problems visible within the agreed time limit.
- Indicative allocation: \$4.0m including software share, implementation, warehouse devices/infrastructure and training.

4.2.4 Initiative 4 — API integration, master data and monitoring

Objective: replace unreliable system connections with secure APIs and create agreed definitions for product, location, price, customer, stock and order data.

- Create standard data formats, an integration platform, automatic retries, data matching, monitoring, alerts, audit records and safe test environments.
- Assign data owners and create quality checks, issue handling, retention rules and cutover checks.
- Protect APIs with access controls, usage limits, input checks and monitoring in line with OWASP API Security guidance.
- Timeline: build the foundation in Months 2–9; continue system connections through Month 29; transfer monitoring and support ownership by Month 30.
- Indicative allocation: \$2.4m across integration, data migration, monitoring and governance.

4.2.5 Initiative 5 — POS and store-fulfilment enablement

Objective: give stores the same product, price, customer, order and stock information, while protecting normal in-store service.

- Connect or selectively modernise POS systems. Support order search, digital receipts, ordering from another location, collection handover, returns and stock adjustments.
- Introduce store task queues, handheld scanning, workload limits, secure staging and proof of collection.
- Pilot in representative stores selected by volume, format, location and change readiness; scale only after accuracy, customer and labour thresholds pass.
- Indicative allocation: \$2.0m for integration, devices, store process changes and staged rollout.

4.2.6 Initiative 6 — Security, privacy, resilience and change

Objective: include security, privacy, recovery and employee adoption from the beginning of every initiative.

- Collect only required customer data, record consent, control access, define retention periods and complete a privacy impact assessment.
- Use the NIST Cybersecurity Framework for security and recovery. Apply PCI DSS controls to the website payment process.
- Provide training by job role, local super users, practice sessions, on-site support, user guides and post-go-live help.
- Indicative allocation: \$1.6m for assurance, testing, resilience, training and change interventions.

4.2.7 Initiative 7 — CRM, loyalty and service integration

Objective: create one customer profile with clear consent controls. Customer-service staff will be able to see preferences, communications, purchases, orders, service cases and loyalty information. Personalised offers will begin with simple rules. Advanced models will wait until privacy controls and customer data are reliable. Indicative allocation: \$1.5m.

4.2.8 Initiative 8 — BI, demand forecasting and personalisation

Objective: use reliable business data to improve decisions. Phase 1 creates starting measures and data-quality reports. Phase 2 adds order, stock and customer dashboards. Phase 3 introduces demand forecasts, suggested replenishment and measured personalisation. Human planners will approve important stock decisions until the models are proven accurate and practical. Indicative allocation: \$1.6m.

5 Implementation phases

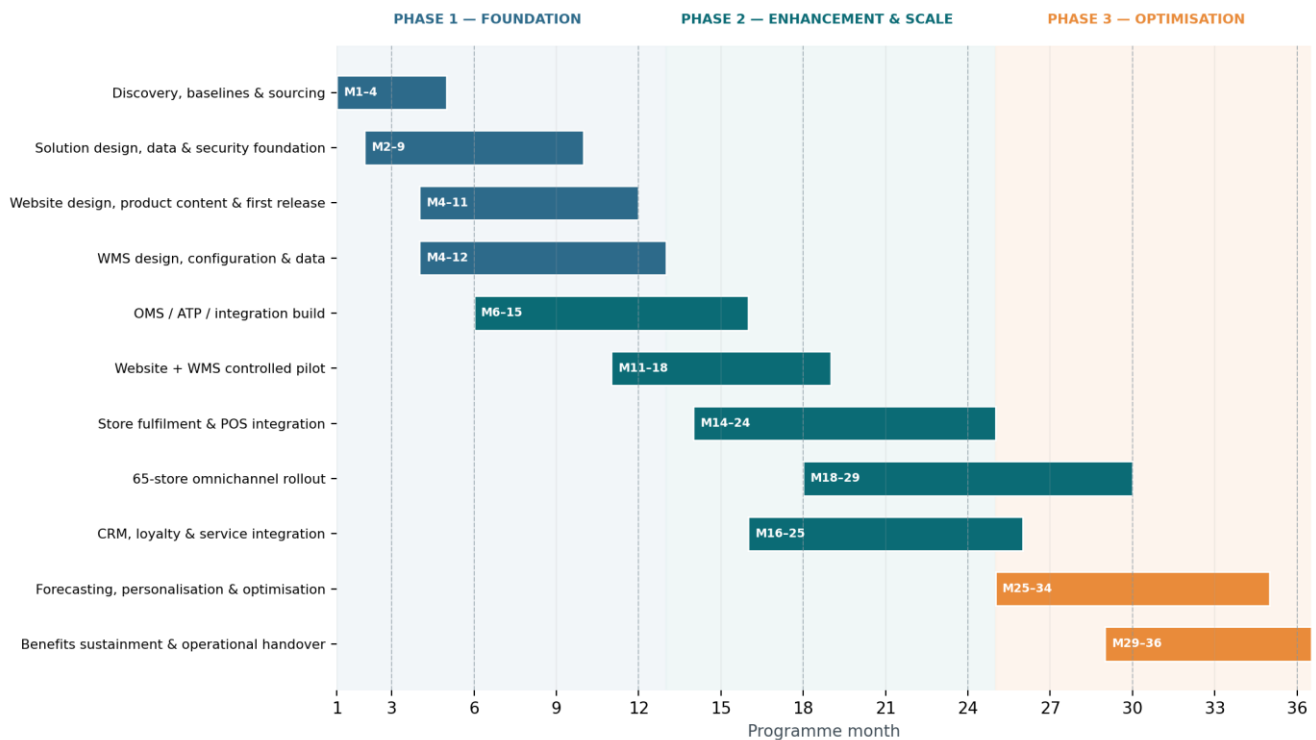


Figure 3. Integrated 36-month roadmap. Timing is indicative and will be refined in the detailed Implementation Plan.

5.1 Phase 1 — Foundation (Months 1–12)

Purpose: confirm the requirements, reduce uncertainty and prepare the website, WMS and stock-visibility pilots without showing unreliable promises to customers.

Workstream	Key activities	Milestones / exit criteria
Governance & discovery	Appoint the sponsor and programme team; measure the starting position; map customer, store and warehouse processes; confirm scope, benefits and risks.	Month 1 mobilisation; Month 3 current-state and baseline approval.
Sourcing & solution design	Issue requirements; evaluate vendors; confirm the system design, performance and security requirements, five-year cost and contract terms.	Month 4 shortlist; Month 6 solution and investment approval.
Data & integration	Assign data owners; check and clean data; build secure API connections, monitoring and data checks.	Month 6 agreed data structures; Month 9 connections and monitoring proven.
First website release	Complete customer research, website structure and design; build content, search, product pages, checkout, payment and reporting.	Month 6 prototype approved; Month 11 release candidate.
WMS pilot preparation	Observe and redesign warehouse work; configure the WMS; prepare devices, network, data, connections, training and cutover practice.	Month 9 process simulation; Month 12 pilot-readiness approval.
Risk & change	Review privacy and security; confirm payment and accessibility controls; assess job changes; design training and support.	No critical control gaps; at least 80% of pilot users trained.

5.2 Phase 2 — Enhancement and scale (Months 13–24)

Purpose: launch the new website, stabilise WMS-controlled warehouse work and introduce connected online/store services through controlled rollout stages.

- Launch the website gradually, keep the old website available as a fallback, monitor performance and provide customer support.
- Go live with WMS in the selected pilot warehouse/DC, stabilise receiving through dispatch, then extend to the remaining confirmed warehouse network.
- Activate the OMS, ATP calculation, stock reservations and warehouse availability. Add store stock only after local accuracy checks are working.
- Pilot click-and-collect and cross-channel returns in representative stores; deploy store task, staging, proof-of-collection and exception processes.
- Integrate CRM/customer service, loyalty foundations, digital receipts and unified order lookup.
- Roll out store capabilities in stages. Pause the rollout if order accuracy, stock accuracy, customer service or system incidents fall outside agreed limits.

5.3 Phase 3 — Optimisation (Months 25–36)

Purpose: complete enterprise adoption and use trusted data to optimise growth, inventory and service.

- Complete the connected online/store rollout across all 65 stores. Remove old system connections only after data checks and audit.

- Introduce selective ship-from-store where item, store, capacity, margin and service rules support it.
- Improve search results, checkout, product recommendations and customer messages through controlled tests.
- Introduce demand forecasting and replenishment alerts. Keep human approval until the forecast model is proven reliable.
- Use WMS data to improve product locations, picking groups, replenishment, labour planning and return decisions.
- Move responsibility to permanent website, warehouse-system and data-support teams. Close the programme only after benefit owners accept responsibility.

6 Risk assessment and mitigation

These risk ratings are high-level estimates for the roadmap. The Implementation Plan will add numeric scores, action dates, owners and escalation rules. A rollout must stop if important controls for customer promises, safety, privacy, payments or business continuity are not ready.

Risk Category	Risk Description	Probability	Impact	Mitigation Strategy	Owner
Data	Incorrect stock, product or price data reaches the website.	High	Very high	Assign data owners; clean and compare data; keep safety stock; create automatic alerts; hide locations or products with unreliable data.	Chief Data Officer
Integration	Delayed or failed system messages show old stock or order status.	High	Very high	Use unique message IDs, automatic retries, an error queue, safe replay, complete monitoring and a customer-safe fallback.	Integration Lead
Operational	WMS go-live interrupts receiving or order fulfilment.	Medium	Very high	Practise with realistic volumes, freeze changes, prepare manual backup work, keep a rollback plan, provide extra support and avoid peak season.	Supply Chain Director
Customer	Click-and-collect promises exceed store capacity or accuracy.	Medium	High	Pilot, capacity limits, ready-time buffers, staging controls, proof of collection and pause rules.	Retail Operations
Cyber/payment	The larger website and third-party payment scripts increase security risk.	Medium	Very high	Apply PCI DSS payment controls, keep an approved script list, develop securely, test regularly, monitor threats and practise incident response.	CISO
Privacy	Customer data is collected or used without a valid reason or consent.	Medium	Very high	Complete a privacy review; collect only needed data; manage consent, retention and access; provide customer opt-out.	Privacy Officer
Adoption	Employees avoid scanning or store tasks, reducing stock accuracy.	High	High	Design work with users, train by role, use local super users, provide on-site support, monitor adoption and simplify difficult steps.	Change Lead
Financial	Scope, system-connection or licence costs exceed the \$20m limit.	Medium	High	Approve funding by stage, use fixed deliverables where suitable, confirm five-year costs, protect emergency funds and control scope.	CFO
Vendor	Dependence on one vendor or a vendor roadmap change limits future options.	Medium	High	Include data-transfer rights, an exit plan and system-connection standards in modular contracts.	Commercial Lead
Delivery	Website, OMS and WMS work combines into one high-risk big-bang launch.	Medium	Very high	Maintain one dependency plan but release each system separately. Use realistic testing, phased pilots and clear stop criteria.	Programme Director
Benefits	Channel shift is counted as growth or savings are not realised.	High	High	Finance-owned baseline, benefit formulas, control groups, capacity removal plans and quarterly validation.	CFO / Benefit Owners
Market	Customer demand, supplier constraints or competitor action changes priorities.	Medium	Medium	Quarterly portfolio review, modular backlog and flexible inventory/promotion rules.	CEO / CMO

6.1 Resilience and rollback principles

1. Protect customer promises: if ATP stock information becomes unreliable, show a safer quantity or temporarily remove the affected delivery option.
2. Release systems separately: the website, OMS rules and WMS should be deployable and reversible independently where possible.
3. Protect transactions: prevent duplicate messages, compare totals, keep order and stock audit records, and replay failed messages in a controlled way.
4. Roll out gradually: start with trained employees, then a selected warehouse and stores, followed by small customer groups and wider rollout stages.
5. Maintain backup processes: keep old lookup and manual fulfilment steps, a payment fallback, tested restore targets and an up-to-date contact list.

6.2 Quality and service standards

Area	Minimum standard
Availability	Target at least 99.95% monthly website availability; use planned maintenance periods; keep key customer functions working if a non-critical service fails.
Performance	Set website page-speed and search/ATP response targets. Test them with expected peak customer volumes.
Security	Review threats, develop securely, scan software components, complete penetration testing, confirm payment controls and practise incident response.
Privacy	Record why data is collected and customer consent; collect only needed data; control retention, access, correction and breach response.
Data integrity	Compare order and stock totals, record data sources, approve master data and monitor missing system messages.
Recoverability	Define how quickly systems and data must be restored; assign rollback authority; test recovery and business-continuity procedures.

7 Investment and ROI analysis

The financial model is a planning estimate, not a vendor quotation. It separates the three-year implementation cost from the operating cost in Years 4–5. All benefits must be checked through discovery, pilots and Finance-approved evidence. Revenue growth is counted only through the additional gross margin it creates, not as total sales revenue. This avoids overstating the benefit.

7.1 Three-year investment breakdown

Investment category	Year 1	Year 2	Year 3	Total
Software licences / subscriptions	\$1.2m	\$1.5m	\$1.6m	\$4.3m
Hardware and infrastructure	\$0.9m	\$1.3m	\$0.4m	\$2.6m
Implementation and integration services	\$2.3m	\$2.9m	\$1.2m	\$6.4m
Data migration, testing and security	\$0.6m	\$0.8m	\$0.3m	\$1.7m
Training and change management	\$0.4m	\$0.6m	\$0.3m	\$1.3m
Internal programme resources	\$0.5m	\$0.6m	\$0.3m	\$1.4m
Contingency reserve	\$0.3m	\$0.4m	\$0.2m	\$0.9m
Total investment	\$6.2m	\$8.1m	\$4.3m	\$18.6m

7.2 Benefits projection

Benefit Category	Year 1	Year 2	Year 3	Total
Revenue growth benefits	\$0.6m	\$3.2m	\$7.2m	\$11.0m
Inventory cost savings	\$0.1m	\$0.8m	\$1.8m	\$2.7m
Warehouse efficiency gains	\$0.1m	\$0.7m	\$1.5m	\$2.3m
Service, reporting and marketing efficiency	\$0.4m	\$1.8m	\$3.1m	\$5.3m
Total Benefits	\$1.2m	\$6.5m	\$13.6m	\$21.3m

The five-year view extends the same benefits into Years 4–5 and includes annual operating costs of \$2.4m. It shows longer-term payback and value while keeping the required three-year projection above.

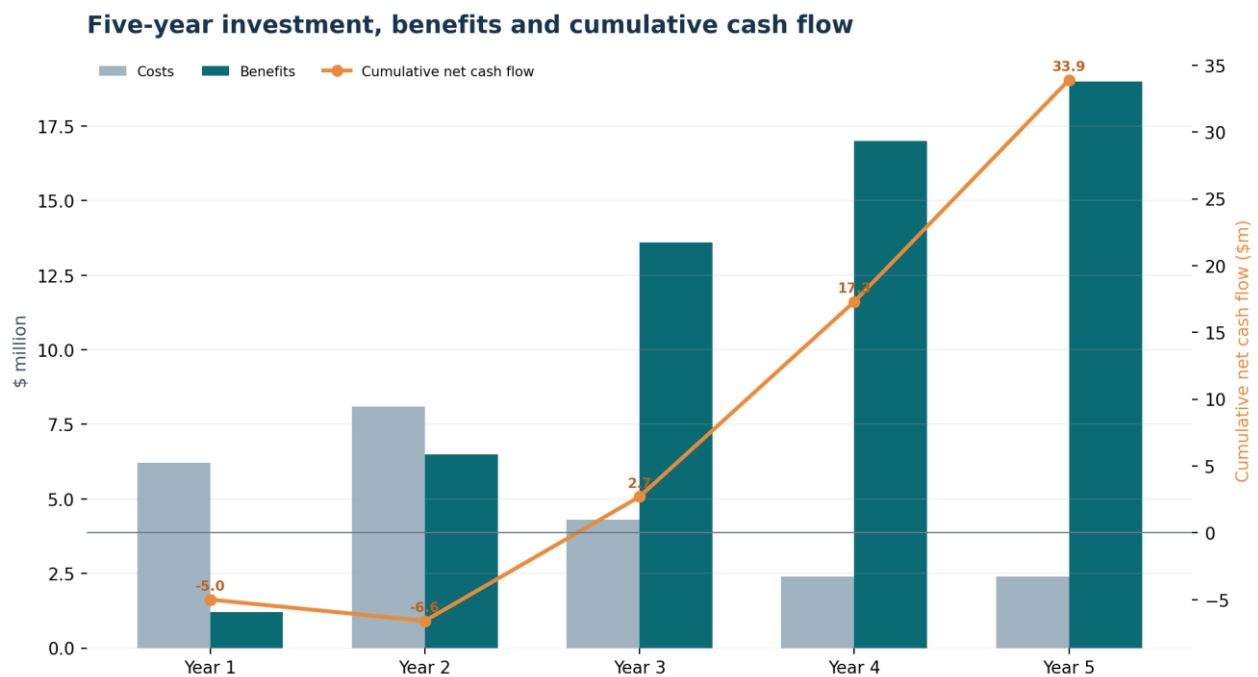


Figure 4. Base-case investment and benefit profile. Years 4–5 include \$2.4m annual operating cost.

7.3 ROI calculation and interpretation

Measure	Calculation	Base result
Three-year ROI	$(\$21.3\text{m benefits} - \$18.6\text{m investment}) \div \18.6m	14.5%
Five-year ROI	$(\$57.3\text{m benefits} - \$23.4\text{m total cost}) \div \23.4m	144.9%
Payback	Cumulative net cash flow becomes positive during Year 3	Approximately Month 33
Five-year net benefit	$\$57.3\text{m benefits} - \23.4m costs	\$33.9m